How curled directions open, three and four: exact walls and pre-registered opening tests in a graph toy with six and eight links per vertex

Emily Smith[1,*]

[1] *Independent researcher, Charlottesville, Virginia, USA*

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We ask how a torus with several curled directions opens, in a graph toy with six and eight links per vertex. The toy is the energy of combinatorial quantum gravity (CQG) with the coefficient λ of its local term raised above 1, later with a tie between directions; only $\lambda = 1$ is CQG. Exact: each curled direction costs $4(\lambda - 1)$ per vertex; each curled state has its own cheapest move out, its wall; the walls rise as directions open, so the energy ties no direction to another; by a published theorem whose proof we have not read, a fully curled start larger than one hypercube is a gas of pieces. Measured, in thirteen pre-registered tests: the first move's price is fixed with size; in no six- or eight-link setting did most runs open every curled direction into one connected region, from one push or in a warm bath. A gas opens cube by cube; a connected torus usually opens one direction or none; full openings were minorities, beside damage, and by shape about a third of them are the start glued together differently. The author's predictions that one push or a warm bath opens every direction failed, as did several of ours. She then chose a tie that releases everything at a vertex's last opening. Under it, in a warm bath, her slab stayed shut or broke, and the untied control showed no difference. The published six-link curve is not reproduced, so every result concerns our family only.

I. INTRODUCTION

Combinatorial quantum gravity (CQG) [1, 2] gives a regular graph an energy built from its squares (4-cycles). A companion paper [3] raised the coefficient λ of that energy's local term above 1 and studied a four-link torus with one direction curled. It found activated escape and a rapid change of the square counts, often into one dominant connected region; the runs that finish clean end as the tube re-glued two along its seam, a flat torus six around where it was four, not two large directions. Only $\lambda = 1$ is CQG. Everything here is at $\lambda > 1$, some of it with an added tie between directions. We call it our family, or the toy: next to CQG and not in it.

The hypothesis's claim (the author's [4]). The curled arrangement that precedes space, which she calls X, has three or four directions curled together; they open together; and one activation releases everything. With four links the claim cannot be posed: a fully curled state, a gas of 4-cubes, has no wall at any $\lambda > 1$ and falls apart at once (O40). With six and eight links it can. This paper poses it twice: exactly, by pricing the single moves out of every rung of the ladder of curled states; and by runs registered in advance, with the author's prediction and ours recorded before any result was read.

Statements are of four kinds and are labeled: *exact* (arithmetic or complete enumeration); *measured* (a pre-registered run, with its registered verdict in capitals and whose prediction held or failed); *interpretation* (ours, unverified); and *the hypothesis's claim* (the author's). "Ours" marks predictions and readings drafted with an AI assistant. Nothing here has been reviewed by a physicist. Tags such as T39 and O81 name entries of the project's pre-registration and assumptions files [4].

Three caveats govern everything below. First, the published six-link curve of CQG [5] is not reproduced by our kernel (Sec. V), and the eight-link kernel is checked against no published curve: the published simulations we rely on are for two and three directions [1, 5]. The author decided to proceed with that gate open. Every six- and eight-link statement is therefore about our family and its kernel, not about the published model. The kernel is the four-link code generalized to any number of links. At four links it reproduces the companion paper's chain random draw by random draw, and its energy and moves are checked by brute force against an independent implementation, at four and six links only. Second, the toy has no time: which direction of a torus would be time it cannot say. Third, "curled" and "open" below are read from counts of squares (Sec. II), not from measured geometry.

Published continuum work treats a change in the number of large dimensions as a decay by tunneling or nucleation [6–8]; there the compact directions change together by construction (a single volume modulus, or a sphere). A fourth paper considers one compact direction opening from a parent with one fewer large dimension [9]. These are benchmarks for what a toy of this kind would have to reproduce, not support for any result here. Their texts were searched by an AI assistant, and no quotation has been checked against the published versions.

II. DEFINITIONS AND EXACT RESULTS

A. The energy, the ladder and the counts

With D directions a vertex has $2D$ links. Graphs are $2D$-regular and bipartite, and no two vertices share more than two neighbors (the hard-core rule, as we read it).

* emilysmithcreate@gmail.com

With S squares, S_e squares on link e, and X the surplus squares above the $2D-2$ that a link carries on the flat torus,

$$H = 16\Big(\tfrac{D(D-1)}{2}N - S\Big) + 4\lambda X, \qquad X = \textstyle\sum_e \big(S_e - (2D-2)\big)_+, \tag{1}$$

derived by us from the published definitions as the four-link energy was (Q21), and not validated against a published curve. The first term vanishes on the flat torus, which has one square per vertex for each of its $D(D-1)/2$ pairs of directions.

Exact (O41, O50). The flat D-torus, every side at least 6, has $H = 0$. A side of length 4 is a *curled* direction: the four links around it close a square, which puts one extra square on every link along it and costs $a \equiv 4(\lambda-1)$ per vertex. The cost is additive, so a torus with k curled directions, the kth *rung* of the ladder, lies ka per vertex above flat. The six-link torus with every side 4 is the 6-cube (64 vertices); its eight-link counterpart is the 8-cube (256).

The count $d(v)$ is the number of pairs of links at v that close no square. On the exact tori it equals the number of directions that are not curled: D on the flat torus, one less per curled direction. Where squares are missing it exceeds D. We call a vertex with $d = D$ *open* and one with $d > D$ *damaged*, and we say a direction *opens* at a vertex where d rises by one. An *open region* is a connected set of open vertices at one reading. On a general graph these counts do not certify a dimension, a product-torus geometry, a macroscopic period or a moving front [4].

B. The walls

The chain's move is a *switch*: it replaces two links by two others and keeps every vertex's number of links. A *wall* is the cost, $-16\,\Delta S + 4\lambda\,\Delta X$, of the cheapest single switch out of an exact torus. A state is *stuck for now* where its wall is positive, which is a property of the energy alone. *Exact* (O49, O50): every switch out of each curled six-link torus was listed and priced; with eight links, and for flat six-link space, only switches with nearby partners were listed, a shortcut checked against the full list on the $4\times4\times18$ torus. Table I gives the cheapest kind at $\lambda = 1.25$. Four things follow, all exact.

(i) The walls rise as directions open. With six links they are $96-80\lambda$, $96-64\lambda$ and $96-48\lambda$ out of the rungs with three, two and one directions curled; with eight, $160-128\lambda$, $160-112\lambda$, $160-96\lambda$ and $160-64\lambda$. Each exceeds the one before by 16λ or 32λ, so the doubling 20, 40, 80 at eight links holds at $\lambda = 1.25$ only (O55). The energy ties no direction to another: each has its own wall.

(ii) The cheapest move out of flat space costs 32, 64 and 128 with four, six and eight links, at every λ (O50, O51). Its ratio to the last rung's wall at $\lambda = 1.25$ is $32/12 = 2.7$, $64/36 = 1.8$ and $128/80 = 1.6$: more directions leave less room between paying the last wall and breaking flat space (O65).

(iii) The fully curled gas is stuck for now only where its wall is positive: $1 < \lambda < 1.2$ with six links and $1 < \lambda < 1.25$ with eight (O41, O50).

(iv) At larger λ another kind of move can undercut the one in Table I. On the two-curled six-link torus it is $128-88\lambda$ (4.8 at $\lambda = 1.40$), so that torus is stuck for $1 < \lambda < 1.45$ (O100); on the three-curled eight-link torus it is $192-136\lambda$ (1.6 at 1.40; T48). A short open side also changes the wall: the $4\times4\times6$ torus admits a move at $96-80\lambda$ that a long open side does not (O49).

Walls do not settle whether a bath warm enough to pay a later wall leaves flat space intact. In an exploratory scan at $\lambda = 1.10$ and 1.25 the flat $6\times6\times6$ torus keeps all 12 squares per vertex up to $g = 6$ and is melted at $g = 10$ (5.1 squares per vertex), with no coupling between at which it rearranges and stays ordered (O51).

C. A fully curled start is a gas

Exact, applying a published theorem [10]; its statement was checked in the abstract and in later citations, and its proof has not been read by us (to verify) (O85). A vertex with every direction curled ($d = 0$) has every two of its links on a square, so any two vertices with a common neighbor have a second one; the hard-core rule allows no third. An arrangement curled everywhere is therefore a $(0,2)$-graph: any two vertices have two common neighbors or none. A connected $(0,2)$-graph of degree k has at most 2^k vertices, with equality only for the k-cube. With $2D$ links that is 4^D: 64 at six links and 256 at eight. So a fully curled start larger than one hypercube is a gas of separate pieces. The gas starts below are not a choice of ours.

TABLE I. Exact: the cheapest single move out of each state at $\lambda = 1.25$, with its cost as a function of λ. Six-link tori $4\times4\times L$ ($L \geq 18$) and $4\times L\times L'$; eight-link tori at $N = 2304$. At larger λ another kind can be cheaper (text).

links	curled	$(\Delta S, \Delta X)$	cost	at 1.25
4	one (the tube)	$(-2,-4)$	$32-16\lambda$	12
4	flat sheet	$(-2,0)$	32	32
6	one	$(-6,-12)$	$96-48\lambda$	36
6	two	$(-6,-16)$	$96-64\lambda$	16
6	three (6-cube, gas)	$(-6,-20)$	$96-80\lambda$	-4
6	flat 3-torus	$(-4,0)$	64	64
8	one	$(-10,-16)$	$160-64\lambda$	80
8	two	$(-10,-24)$	$160-96\lambda$	40
8	three	$(-10,-28)$	$160-112\lambda$	20
8	four (8-cube, gas)	$(-10,-32)$	$160-128\lambda$	0
8	flat 4-torus	$(-8,0)$	128	128

D. Ties

A *tie* adds $T_f = \sum_v f(d(v))$ to H, with $f(0) = f(D) = 0$ and $f = 0$ at damaged vertices. It is a rule put in by hand, at the author's decision [4], and the toy with it is our family. *Exact*, six links (O70): a state with d directions open lies $(3-d)a + f(d)$ per vertex above flat, so the gaps between rungs become $a - f(1)$, $a + f(1) - f(2)$ and $a + f(2)$ in order of opening; their sum stays $3a$.

Two forms matter below. The *follow* form is $f(d) = \kappa(D - d)$ for $1 \le d \le D$. At $\lambda = 1.25$ its walls out of the gas, the two-curled torus and the one-curled torus are -4, 16, 36 at $\kappa = 0$; 20, -6, 20 at $\kappa = 1$; 44, -30, 4 at $\kappa = 2$; and 56, -42, -4 at $\kappa = 2.5$; flat space's wall stays 64 (O68, O71). It makes the gas stuck and removes the walls of the partly open rungs, the middle one first. The form *all at the last* is $f = (0, a, 2a, 0)$ for $d = 0$ to 3, with no constant to fit. Every rung then lies $3a$ per vertex above flat, so nothing is gained until a vertex's last direction opens (O89). Table II gives its walls (O106).

III. MEASURED: EVERY ATTEMPT TO OPEN MORE THAN ONE DIRECTION

Three protocols are used. *Sealed*: a bath of C stores (a Creutz reservoir of C demons), one holding the push and the rest empty; a proposed switch draws on one store at random and is refused if that store cannot pay; H, the tie where there is one, and the stores are conserved together. *Local heat*: sealed, with one store per vertex, so released energy stays with the vertex that releases it. *Warm bath*: the Metropolis chain at a fixed coupling g, with no push. Conservation does not establish equilibrium, and a mean store energy is not a calibrated temperature [4]. The sealed dynamics are those of the companion papers [3, 11].

Table III lists every test. Each registration defines its per-replica outcome words from the number of vertices at each value of d. In brief, at a run's last reading: STUCK or STAYS, most vertices still on the starting rung; ADVANCES, the most populated rung is above the start; OPENS (T48, T53), at least half the vertices open and the largest open region holding at least half of all vertices; MELTED or DAMAGED, a quarter or more of the vertices damaged. A cell (one start at one setting) takes the outcome of most of its replicas, and the verdict follows from the cells by the registered rule.

TABLE II. Exact (O106): single-move walls under the tie all at the last, six links. The rod is $4 \times 4 \times 18$ (two curled); the slab is $4 \times L \times L$ (one curled; $L = 8$ and 12 alike). Untied walls are in Table I.

λ	a	$3a$	gas	rod	slab	flat
1.05	0.2	0.6	14.4	30.4	39.2	64
1.10	0.4	1.2	12.8	28.8	30.4	64
1.15	0.6	1.8	11.2	27.2	21.6	64
1.20	0.8	2.4	9.6	25.6	12.8	64
1.25	1.0	3.0	8.0	24.0	4.0	64

A. The price of the first move is fixed with size

Measured (T32). A single store held the push, at $N = 192$, 288, 384 and 512, ten replicas per push, 20,000 sweeps. No replica left the start at 12, 14 or 15. Every replica left at 16 and above, at every size: FIXED WALL, sharp. A replica counts as leaving when its squares change at all, so this is the price of the first move, the wall of Table I. It is not the price of a conversion, as the next tests show.

Interpretation (ours, unverified; O103). On the four-link tube and on the $4 \times 4 \times L$ torus the cross-section is fixed, 4 and 16 vertices, so the boundary between converted and unconverted parts cannot grow. On a $4 \times L \times L$ torus it grows with the converted patch. A first-move price that is fixed with size is therefore not evidence that the barrier to opening a slab is fixed (Sec. IV).

B. One push, untied

Measured (T30; O54, O64). At $\lambda = 1.25$, push 16, twelve replicas per cell and 100,000 sweeps, 84 replicas in seven cells. By the final count the first curled direction opened completely, no vertex left at $d = 1$, in 28. In 21 it went most of the way (15 to 24% of vertices still at $d = 1$). In 32 it stalled early (36 to 53% still at $d = 1$; energy 1.86 to 1.95 per vertex against 2.0 at the start). In 3, at the hottest baths, it went past the one-curled rung; at $C = N/8$ two of twelve replicas reached a defective near-flat state (88 to 90% of vertices open), just under the registered bar. At $\lambda = 1.10$ a push of 26 (the wall is 25.6) bought one move, which neither healed nor grew. No torus cell has a majority that is flat or on the one-curled rung: NEVER OPENS. The second direction never opened in a cold bath.

The gas of eight 6-cubes at $\lambda = 1.10$ is scored separately (T30-gas). In 21 of 24 replicas the energy fell from 1.2 to between 0.24 and 0.41 per vertex and about 40% of vertices ended open, in patches whose largest held 36 to 64 vertices; one cube holds 64. None rested on the level with one direction open for the registered 5,000 sweeps. Three did not leave. By the letter the verdict is FIRST ONLY, which for the gas means two of three directions open. It was never one space.

Measured (T33; O62). With eight links at $N = 2304$ each start was given a push of 20, its wall (19.2 for the gas). In every replica the push bought one move, the energy rose by that move's cost (3.000 to 3.009 and 1.600 to 1.608 per vertex), the bath was left empty and nothing changed in 50,000 sweeps. NO CASCADE; no pattern was tested, because nothing opened.

Measured (T40, T39, T41; O81). Larger pushes, 20 to 160, did not start four directions: NEVER STARTS from the three-curled torus and from a gas of four 8-cubes, with no replica in which most vertices left the starting rung. Scanning the bath size found no window in which the first release pays the next wall and the space survives: NO WINDOW in all four rows. With six links most replicas at $\lambda = 1.25$ ended partly converted with no flat majority at any bath, and at 1.40 the smaller baths read MELTED; with eight links every cell read STUCK. The simplest three-direction case, one curled direction given its wall (36 and 28.8), stayed curled in all 144 replicas over 100,000 sweeps: NEVER OPENS.

C. One push, with a tie

Measured (T44; O79). Under the follow tie the gas of eight 6-cubes was given its tied wall (20, 44, 56) in baths of $N/2$ and $2N$ stores. All 48 replicas read STUCK: 97% of vertices were still at $d = 0$, and the push made one patch of about 16 vertices. NOT ALL at every κ.

Measured (T45, T46; O79). A tie with two constants was run in four settings on the same gas, and one with three constants in two settings on a gas of nine 8-cubes, each with two pushes: the first wall alone, and the sum of the positive walls. Their constants had been fixed by a target ratio of releases that the record has since withdrawn [4]; the runs stand as run. NOT ALL in every setting. With six links the setting $(f(1), f(2)) = (0.528a, -a)$ at $\lambda = 1.25$ rested with two directions open and the last curled, where its exact wall of 52 put it; the other settings read MELTED or PARTLY OPEN by cell. With eight links all six replicas given the first wall (17.61) read STUCK, 99% of vertices still at $d = 0$, and all six given the sum (about 174) read MELTED: 37 to 38% of vertices at $d = 2$, 23 to 24% at $d = 3$, 2.5 to 2.7% at $d = 4$ and 26% damaged.

D. What the saved wiring shows

A retrospective reading of saved final graphs, not preregistered; it changes no verdict (O85; 156 graphs of T44 to T46). From a gas, no open region exceeded one starting piece: at most 64 of 512 vertices with six links and at most 4 of 2304 with eight. One open region in 156 graphs held vertices of two pieces. In one T45 setting at $\lambda = 1.40$, 10 cubes in 24 graphs opened completely on their own. The damaged vertices are not disordered blobs: they lie on links between starting pieces and on boundaries between open and curled parts. In that setting 50 to 81% of them are ends of links between two starting pieces; in the other settings, and in the connected torus of T39, 33 to 69% touch both an open and a curled vertex; nowhere do more than 5% have only damaged neighbors. So the registered word MELTED in these runs does not mean a melted space. Each cube opens, or partly opens, inside itself, and the cubes do not join into one space (O85, O100). The graphs of T30-gas were not among those read; its largest open patches are no larger than one cube.

Two consequences. The share of open vertices cannot tell one large region from many small ones; the largest open region can, and it is recorded at every reading from T48 on. And a connected start differs: from the $4\times4\times18$ torus (T39, $\lambda = 1.25$, $C = N/16$) the largest open region reached 186 to 250 of 288 vertices in 9 of 12 replicas, with a damaged boundary beside it.

E. Released energy kept in place, and one cube alone

Measured (T48; O91). With local heat and no tie, the four-link control read ONE SPACE at $\lambda = 1.25$ (15 of 16 replicas open) and ADVANCES ONLY at 1.40 (5 of 16). The two-curled six-link torus read ADVANCES ONLY at 1.25: 13 of 16 replicas rest with one more direction open and the last curled. At 1.40 it read DAMAGED by the letter. The one-curled torus read STAYS in 32 of 32. The three-curled eight-link torus read STAYS at 1.25 and ADVANCES ONLY at 1.40, where 12 of 12 rest one rung up.

Not scored, and the result closest to the claim: at $\lambda = 1.40$ the two-curled torus met the OPENS rule in 5 replicas of 16, every curled direction open in one region holding at least half the vertices, reached between 45,000 and 197,000 sweeps (4 of them nearer flat space by shape; Sec. IV A); 7 were damaged. At that λ the start's wall is 4.8 and each rung's gap is 1.6 per vertex. It is a minority, beside damage, at one size.

Measured (T49; O92). One hypercube alone, 32 replicas per cell, with named points and with interchangeable points (unlabeled graphs: relative to named points, each state is weighted by its number of symmetries). The 6-cube where it is stuck ($\lambda = 1.10$ and 1.15, pushes 8 and 4): named points read PART OPEN (31 and 27 of 32; 1 and 5 fully open); interchangeable points read STUCK (20 of 32) and MIXED. Where it has a move downhill ($\lambda = 1.25$, no push): PART OPEN with both, fully open in 4 of 32 named and 14 of 32 interchangeable. The 8-cube read STUCK, 32 of 32, at $\lambda = 1.10$ and 1.15 with either kind of point, and PART OPEN at 1.30 (27 of 32; 5 damaged). The shared prediction held where the cube is stuck, except that the stuck 8-cube did not move at all, and failed where the cube has a move downhill: it does not open fully. *Interpretation (ours, unverified):* a move downhill out of a cube is not a path down to the open state.

F. A warm bath

Measured (T53; O105). The two-curled torus was held in a warm bath, untied, eight replicas per cell, 200,000

sweeps, at $N = 288$, 576 and 1152. The first curled direction opened at every length and both λ; one cell, the coldest bath at $L = 18$ and $\lambda = 1.25$, read STAYS (5 of 8). No cell has an OPENS majority. At $\lambda = 1.25$ no replica opened its second direction at any length or bath, and at $L = 72$ none did at either λ. At $\lambda = 1.40$ the second direction opened in minorities: 2, 1 and 4 of 8 at $g = 2.0$, 2.5 and 3.0 at $L = 18$, and 1 of 8 at $g = 2.5$ and 3.0 at $L = 36$, in cells that read MIXED or DAMAGED. The verdict is ADVANCES ONLY at all six (L, λ). The registered count of separate regions early in a run has median 2 at $L = 72$ and 1 at $L = 18$: SEVERAL.

Interpretation (ours, unverified; O105). The bath that crosses the second wall (28.8 at $\lambda = 1.40$) also crosses flat space's wall of 64. Size works against the second opening: it was seen at $L = 18$ and 36 and never at 72.

IV. THE TIE THE AUTHOR CHOSE, AND ITS FIRST TEST

The hypothesis's claim (the author's, 9 October [4]). Her decision: the directions are tied so that nothing is released until a vertex's last direction opens, all at the last; untied runs are a stepping stone and serve as controls. Her statement, not a decision: the stand-in for the curled start that matches her picture is the slab, $4\times L\times L$, two directions open and one curled.

Exact (O89, O106; Table II). Under this tie the slab's wall falls from 39.2 at $\lambda = 1.05$ to 4.0 at 1.25 while flat space's wall stays 64, and the gap to flat is $3a$, 0.6 to 3.0 per vertex. The gas and the rod lie at the same $3a$ as the slab, so the openings that lead from them to the slab gain nothing, and their walls are 8 to 30. *Interpretation (ours, unverified).* Those earlier openings would have to be carried by a bath. The ratio of the slab's wall to flat space's is 0.34 at $\lambda = 1.15$ and 0.20 at 1.20. Untied at $\lambda = 1.25$ it was $36/64 = 0.56$, and the four-link tube, which converted in a bath at $g = 1.5$ [3], has $12/32 = 0.375$. At $\lambda = 1.25$, with a wall of 4, the slab is not stuck in any bath. Two cautions. The single move is not the whole barrier in three directions. And under the tie as implemented a damaged vertex pays no tie, so a slab vertex is relieved of its $2a$ by breaking as well as by opening.

Exact (T54; O118, as corrected). T54 prices completed patches: the energy $\Delta H(k)$ of a slab whose curled direction is opened over a patch of $k\times k$ columns, wired as the saved end states of T48 are wired. These are energies of completed square patches and not a barrier: going from one patch to the next takes $2k + 1$ switches, and T54 did not price the states between. The registered verdict words are FIXED WALL, CRITICAL PATCH and NO FINITE PATCH, read over completed patches at a held-out size, $L = 16$, after a prediction for it was written from a fit at $L = 8$ and 12. The author's prediction was CRITICAL PATCH, with a small patch as her lean; ours was CRITICAL PATCH, amended before the held-out size to FIXED WALL at $\lambda = 1.40$ under the tie. At all three sizes $\Delta H(k)$ is the same exact quadratic, $\alpha k - \beta k^2$, with nothing left over, for every patch that does not wrap around the torus. Untied, $\alpha = 80-16\lambda$ and $\beta = 16(\lambda-1)$ (read off the values, not derived): at $\lambda = 1.25$ it reads 56, 104, 144, 176 for $k = 1$ to 4, and the fitted curve peaks at $k = \alpha/2\beta = 7.5$ columns, at 225; at $\lambda = 1.10$, at 19.5 columns and 608. Under the tie $\alpha = 176 - 112\lambda$ and $\beta = 48(\lambda - 1)$: the fitted curve peaks at 1.5 columns and 27 at $\lambda = 1.25$, 2.2 and 45 at 1.20, 3.3 and 77 at 1.15, and 5.5 and 145 at 1.10. The tied numbers at $L = 8$ were computed before the author confirmed her tied prediction, as the registration discloses. The prediction for $L = 16$ was written from these, each within 10%; for the untied $\lambda \le 1.25$ the fitted peak lies beyond what $L = 16$ can hold, and those were registered as unreadable there. *The held-out size* ($L = 16$): the quadratic fitted to its own values has exactly the same coefficients, so all six readable numbers hold. The verdict word, read as registered over every k computed, is RISES AGAIN, a word that is reported and not scored, at $\lambda = 1.10$ to 1.20 tied and at 1.40 untied: the last patch, $k = 15$, wraps the torus and lies above the quadratic. It is FIXED WALL at $\lambda = 1.25$ tied, where two whole patches tie at 24, and at 1.40 tied, where the first patch is free. So the author's CRITICAL PATCH is not scored in four rows and fails by the letter in the two that are scored; ours fails at tied 1.25 and holds at tied 1.40. Among the scored rows NO FINITE PATCH does not appear.

Exact (an outside review found this with its own code, and ours reproduces it). The fitted peak is not a barrier and bounds none. Between the first two patches under the tie at $\lambda = 1.25$, which both stand at 24, the best order of the three switches passes through 24, 20, 45, 24: the highest energy met is 45, above the fitted 27. At $\lambda = 1.20$ the first two steps reach 62 and 66 against a fitted 45, and at 1.15, 80 and 98 against 77. Nor are square patches the cheapest shapes. A single line of columns along one of the construction's two directions costs $\alpha - \beta$ for its first column and $96 - 80\lambda$ more for each further one under the tie: 24, 20, 16, 12, 8, 4 at $\lambda = 1.25$, level at 32 at 1.20, and rising by 4 and by 8 a column at 1.15 and 1.10; untied, every further column costs $32 - 16\lambda$. So nothing here shows that the tie turns the climb into one a bath could make. The barrier to opening the slab's curled direction, tied or untied, is not known: it is the minimax barrier (the highest energy on the lowest path) and needs a search over paths with every state priced, shapes other than squares included.

Measured (T56, stage 1; O115). The slab ($L = 8$ and 12; 256 and 576 vertices) and the rod ($L = 18$ and 36) were held in a warm bath under this tie at $\lambda = 1.10$, 1.15, 1.20 and 1.25 and $g = 1.5$ to 3.0, eight replicas per cell, 200,000 sweeps, with four untied slab cells at $\lambda = 1.20$ as the control. No cell of 64 has an OPENS majority. The slab's window reads STAYS at $\lambda = 1.10$, 1.15 and 1.20 and ADVANCES ONLY at 1.25, at both sizes. The rod's reads STAYS at every λ and both lengths. The con-

trol reads NO DIFFERENCE: tied and untied both stay. SEVERAL for both shapes. Of 256 tied slab replicas 2 read OPENS; of 256 tied rod replicas 17, all at $\lambda = 1.20$ or 1.25 in baths of $g = 2.0$ to 3.0, beside damage. None of the 19 ended flat: 0.74 to 2.6 per vertex was left. The author's prediction, that the slab OPENS IN A WINDOW into ONE SPACE, failed at every λ. Ours failed too: a window at 1.15 and 1.20, and that the tie would make the difference. SEVERAL held.

Two things, not scored, say why. First, as the second caution above warns, the tied slab is damaged where the untied one is untouched: 5 to 10% of its vertices on average at $\lambda = 1.20$, between an eighth and three tenths at 1.25 in every bath, and none in the 32 untied replicas. Second, where the slab stayed, opening takes more than the single move, though how much more is not known: T54's completed patches stand 32 to 145 above the slab at $\lambda = 1.20$ to 1.10, the states between them stand higher, and a line of columns costs 32, 40 and 48 for its first column and no less for each further one (O118, as corrected). *Interpretation (ours, unverified):* our prediction read the single-move wall as the barrier, against the first caution. By the registered rule the prediction for the held-out sizes ($L = 16$ and 72) is that nothing opens. That stage has not been run, so nothing here is a claim about size.

A. OPENS by the count, and the shape

A retrospective reading of saved final graphs: exploratory, exact for each graph; it changes no verdict (O116). OPENS counts open vertices in one connected region; it does not say how wide the region is. One rod of T56 ended with every vertex open, none damaged, and 1.2 per vertex above flat space. So for every replica that reads OPENS we counted the vertices at each distance from a vertex. The $4 \times 4 \times 18$ rod has at most 32 at any distance, and its farthest vertex is 13 steps away. The flat $6 \times 6 \times 8$ torus, with the same 288 vertices, has up to 60, and its farthest is 10 away. That rod of T56 has the rod's numbers: it is the starting torus glued together differently, so that no vertex reads as curled. Sorting each replica that reads OPENS by the nearer profile: for T48's two-curled torus at $\lambda = 1.40$, 4 of 5 are nearer flat space; for T53, 7 of 9; for T56's rod, 9 of 17. T56's two slab replicas are a little wider than the slab. In the four-link control of T48 at $\lambda = 1.25$, 9 of the 15 replicas that read OPENS have the tube's profile, eight vertices at each distance, as has the tube re-glued two along its seam; 6 are wider. The companion paper's maps show the same at scale: of their 11,880 runs, 9,865 finished exactly flat, nearly all of them that re-glued tube, six around where it was four. So OPENS and ONE SPACE in Table III are met by the count in every case listed, and by the shape in about two thirds of the six-link ones. A width or a distance profile belongs in any registration that uses those words.

V. THE REPRODUCTION GATE

This project reproduces a published curve before interpreting results built on the same code. The six-link target is Fig. 3 of Ref. [5]: squares per vertex against $\ln \hbar g$ for $D = 3$ at $\lambda = 1$, which is CQG. Its caption gives neither the size of the graphs nor a protocol; we ran $N = 500$, the size the paper states for another figure (O96). *Measured.* Under the plain reading of the axis our curve is ordered at every published coupling (Gate C, failed; O43). A reading in which the weight goes as $1/g^2$ was pre-registered and FAILS, and the same code's two-dimensional figure is not compressed (ONE POWER) (Gate C′; O53). One difference has since been accounted for (O96): the energy of Ref. [5], its Eq. (1), counts every link twice, and ours counts it once, a factor of two in the coupling. With the coupling halved, our cooling legs cross 9 and 6 squares per vertex at 0.677 and 0.737, and at 0.970 and 0.961, in $\ln \hbar g$, against the published 0.666 and 0.996; this was reported after the run and not scored. Two differences remain. The published curve reaches 2 squares per vertex a factor 2.8 sooner in coupling, and its plateau is 10.07 against our 11.4. *Interpretation (ours, unverified).* Two readings are open. The published runs may have been short and out of equilibrium: in an exploratory run a fast heating leg from a slowly cooled state comes near the published plateau and width (O45). Or the published graphs may carry triangles and pentagons, which the paper's text allows and our bipartite kernel does not (O96); against that, the published hot end, 1.09 squares per vertex, sits at our bipartite value, 1.15, where such graphs would sit near half of it. The gate stays open. The exact results of Sec. II are properties of the energy as we derived it and do not depend on the sampler.

VI. DISCUSSION

The summary so far is negative. *Exact:* the ladder is additive; every rung has its own wall and the walls rise as directions open; flat space's wall is 32, 64 and 128 at every λ; and a fully curled start larger than one hypercube is a gas. *Measured:* the price of the first move is fixed with size, and in thirteen pre-registered tests, untied or under the ties run, no push of any size tried, no bath size and no warm bath gave a six- or eight-link setting in which a majority of runs had every curled direction open in one region. A gas opened cube by cube. A two-curled torus opened one direction and rested, or opened fully in a minority at $\lambda = 1.40$, beside damage. A one-curled torus given its wall did not move, and under the author's tie in a warm bath it stayed or broke. The four-link control read ONE SPACE at $\lambda = 1.25$, though by shape most of those had the profile of the tube re-glued six around, no wider (Sec. IV A). The author's predictions that one push, or a bath, opens every direction failed in every test that could score them; her predictions that the first move's price is fixed with size (T32) and that openings

start in several places (T53) held. Several of ours failed too: FIRST ONLY for the tori of T30, the windows we expected at $\lambda = 1.40$ and 1.50 (T39, T41, T53), and the push that starts four directions (T40).

In the project's form for a failed prediction [4]: *previous claim* (the author's), one activation opens every curled direction and releases everything. *Failed because* each direction has its own wall and the energy ties nothing: a push the size of the cheapest move opens one direction, or one cube, and stops (T33, T40, T44 to T49; O103). *Replacement* (the author's): the curled start is one connected piece with a direction already open, and a bath supplies each push in turn. *Its test*, T53, read ADVANCES ONLY: the bath opens the first direction and not the second before it damages the space. *Second replacement* (the author's): the tie all at the last, with the slab as the stand-in. *Its test*, T56, read STAYS or ADVANCES ONLY for the slab, with NO DIFFERENCE from the untied control: an opening that spreads takes more than one move, and the table rewards breaking (O115). A third replacement is the author's to propose; none is written here. *New falsification tests:* T54 to its held-out size under the tie, and the exact price of the cheapest breaking move out of the slab against the cheapest opening one. The branch is abandoned in this family if T54 reads NO FINITE PATCH, or a barrier above flat space's, at every λ at which the slab is stuck; or if a table that also charges damage, chosen by the author and priced exactly first, still does not open it (O115). That condition was written when T54's numbers were taken for a barrier; since they are not one (Sec. IV), deciding it needs a search for the lowest path, not yet designed.

What the curled start may be is the author's to say. Her statements so far [4]: it may begin with one direction already open; a foam of small fully curled pieces that open separately and then join she holds open, not adopted; it may be long in one open direction with two curled; and the shape that matches her picture is the slab. In the toy, only connected starts with a direction already open have reached one large open region, and no run has shown separate pieces joining (O85).

Interpretation (ours, unverified). What stopped the untied openings is the wall of the next direction, which rises as directions open while flat space's wall stays fixed. The tie lowers the slab's single-move wall (Table II against Table I) and puts the whole gap at the last opening. In T56 that was not enough: where the wall was high the slab stayed, and where it was low the slab broke, because the tie as written charges nothing for a broken vertex. A tie is put in by hand: even a positive result would show only that the mechanism is possible in this family with that rule, not that reality has it. The room a conversion needs in two directions [12], and whether concentrated energy curls flat space [13], are treated in companion drafts. No outcome here says anything about the real universe.

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TABLE III. Measured: the pre-registered tests, with the registered verdict as the record gives it. Starts are given as links; state; torus. "Push = wall" means the push equals the exact wall of the start. A: the author's prediction; O: ours; inf.: inferred for her by the AI assistant from her stated positions and not confirmed by her before the reading. Named points throughout, except in T49. Where two values of λ have different verdicts they are listed in order.

Test	Start	λ	Protocol	Registered verdict	Predictions	Source
T32	6; two curled; $4 \times 4 \times L$, $N = 192$ to 512	1.25	sealed, one store, pushes 12 to 20	FIXED WALL (16, sharp)	A held; O held	O54
T30	6; two curled; $4 \times 4 \times 18$, $4 \times 4 \times 32$	1.25, 1.10	sealed, push = wall, $C = N/8$ to $2N$	NEVER OPENS	A (ALL AT ONCE) failed; O (FIRST ONLY) failed	O54, O64
T30-gas	6; gas of eight 6-cubes	1.10	sealed, pushes 8 and 10, $C = 2N$	FIRST ONLY, by the letter	A (ALL AT ONCE) failed	O54, O64
T33	8; three curled, $4 \times 4 \times 4 \times 36$; gas of nine 8-cubes	1.25; 1.10	sealed, push 20, $C = N/8$ to $2N$	NO CASCADE (every replica STALLS)	A, O: not tested, nothing opened	O62
T39	6; two curled, $4 \times 4 \times 18$ (1.25, 1.40). 8; two curled, $4 \times 4 \times 8 \times 8$ (1.25, 1.50)	1.25 to 1.50	sealed, push = wall, $C = N/16$ to $2N$	NO WINDOW (all four rows)	A (inf.: WINDOW) failed; O held at 1.25, failed at 1.40 and 1.50	O81
T40	8; three curled, $4 \times 4 \times 4 \times 12$; gas of four 8-cubes	1.25; 1.10	sealed, pushes 20 to 160, $C = N/2$	NEVER STARTS (both)	A (inf.: a tied pattern) not reached; O (starts at 40 to 80) failed	O81
T41	6; one curled; $4 \times 8 \times 12$	1.25, 1.40	sealed, push = wall, $C = N/8$ to $4N$	NEVER OPENS (both)	A (inf.: ROOM NEEDED) failed; O held at 1.25, failed at 1.40	O81
T44	6; gas of eight 6-cubes	1.25	sealed, follow tie, $\kappa = 1$, 2, 2.5, push = wall	NOT ALL (every κ)	A (inf.: ONE PUSH OPENS ALL) failed; O (NOT ALL) held	O79
T45	6; gas of eight 6-cubes	1.25 to 1.46	sealed, two-constant tie, four settings, two pushes	NOT ALL (all four)	A failed (B inf., C hers); O (none opens all) held	O79
T46	8; gas of nine 8-cubes	1.30	sealed, three-constant tie, two settings, two pushes	NOT ALL (both)	A (ONE PUSH OPENS ALL) failed; O (NOT ALL) held	O79
T48	4; one curled; 4×48 (control)	1.25, 1.40	local heat, push = wall and four times it	ONE SPACE; ADVANCES ONLY	inf. and O (ONE SPACE) held at 1.25, failed at 1.40	O91
T48	6; two curled; $4 \times 4 \times 18$	1.25, 1.40	the same	ADVANCES ONLY; DAMAGED	A (ADVANCES ONLY) held at 1.25, failed at 1.40; O the same	O91
T48	6; one curled; $4 \times 8 \times 12$	1.25, 1.40	the same	STAYS; STAYS	O (STAYS) held; inf. (ONE SPACE) failed	O91
T48	8; three curled; $4 \times 4 \times 4 \times 12$	1.25, 1.40	the same	STAYS; ADVANCES ONLY	A (ADVANCES ONLY) failed at 1.25, held at 1.40; O the same	O91
T49	6; one 6-cube. 8; one 8-cube	1.10 to 1.30	sealed, push = wall, or none where downhill; named and interchangeable points	PART OPEN or STUCK by cell (text)	shared (A adopted O): held where stuck, failed where downhill	O92
T53	6; two curled; $4 \times 4 \times L$, $L = 18$, 36, 72	1.25, 1.40	warm bath, $g = 1.5$ to 3.5, no push	ADVANCES ONLY (all six); SEVERAL	A (OPENS IN A WINDOW) failed; A (SEVERAL) held; O (ADVANCES ONLY at 1.25) held; O (a window at 1.40, $L = 18$; ONE) failed	O105
T56	6; one curled (slab), $4 \times L \times L$, $L = 8$, 12; two curled (rod), $4 \times 4 \times L$, $L = 18$, 36	1.10 to 1.25	warm bath, $g = 1.5$ to 3.0, tie all at the last; untied controls	slab STAYS, ADVANCES ONLY at 1.25; rod STAYS; control NO DIFFERENCE; SEVERAL	A (the slab OPENS IN A WINDOW, ONE SPACE) failed; O (a window at 1.15 and 1.20; the tie makes the difference) failed; O (SEVERAL) held	O115